Basic Math for AI and ML

Mathematics for AI / Machine Learning

AI models learn from **numbers and patterns**.
Mathematics helps the computer **understand data, adjust mistakes, and improve predictions**.

The **3 main math areas used in AI/ML** are:

1. Linear Algebra
2. Calculus
3. Optimization

Think of AI like **teaching a robot to become smarter step by step.**

1. Linear Algebra (Vectors & Matrices)

Simple Idea

Linear algebra helps the computer **store and handle large amounts of data**.

Think of it like **organizing numbers in boxes and tables**.

Example:

Student marks table

Student	Math marks	AI marks
Ali Hamza	80	85
Bushra	81	82
Shakeel	83	80

This table is a matrix.

Vector

Simple Idea

A **vector** is simply a **list of numbers**.

Example: list = [80,40]

This could represent a student's marks.

In AI, a vector can represent:

- an image
- a text sentence
- a customer profile
- a sound signal

Purpose in AI

Vectors help convert **real-world things into numbers**.

AI only understands **numbers**, not images or words.

Example in AI

If we give AI a photo of a cat, it converts it into numbers like this:

```
[255, 0, 120, 90, ...]
```

These numbers form a **vector**.

So:

Vector = Data representation

Matrix

Simple Idea

A **matrix** is a **table of numbers**.

Example

```
[ 1  2  3
  4  5  6
  7  8  9 ]
```

Matrices are used to represent **large datasets**.

Purpose in AI

Matrices help perform **large calculations quickly**. AI models like **Neural Networks** multiply matrices to process data.

Example in AI

Input data × Model weights = Prediction

For example:

`Input data × Weight matrix = Output` (This happens millions of times during training.)

So:

Matrix = Fast data processing

2. Calculus (Derivatives & Gradients)

Simple Idea

Calculus helps AI **learn from mistakes**.

When AI makes a wrong prediction, calculus tells it:

"How should you change yourself to become better?"

Derivative

Simple Idea

A **derivative** tells us **how fast something is changing**.

Example:

Imagine climbing a hill.

Derivative tells you:

- how steep the hill is
- whether you are going up or down

Purpose in AI

Derivatives help AI **measure error changes**.

- They answer this question:
- If we change the model slightly, will the error increase or decrease?
- This helps AI improve predictions.

Example in AI

AI predicts house price:

Real price = 100,000
Predicted = 120,000

Error = 20,000

Derivative tells AI:

Which direction should we adjust the model?

Gradient

Simple Idea

Gradient means **direction of fastest change**.

Imagine you are on a mountain and want to go down quickly.

Gradient tells you:

which direction goes downhill fastest.

Purpose in AI

Gradient helps AI find the **best model settings**.

It tells AI:

Move your parameters in this direction to reduce error.

Example in AI

Used in training:

- Neural Networks
- Deep Learning
- Logistic Regression
- CNN models

So:

Derivative → How error changes
Gradient → Which direction improves the model

3. Optimization

Simple Idea

Optimization means **finding the best solution**.

In AI we want:

Lowest error and best prediction

Example

AI predicts exam marks.

Prediction = 70
Real marks = 90

Error = 20

Optimization tries to **reduce this error**.

Gradient Descent (Most Important)

Simple Idea

Gradient descent is like **walking down a hill step by step** to reach the lowest point.

Top of hill = large error
Bottom of hill = minimum error

AI slowly adjusts its parameters to reach the **lowest error point**.

Purpose in AI

Optimization helps AI:

- reduce prediction errors
- train neural networks
- improve accuracy

Without optimization:

AI **cannot learn properly**.

Simple Real-Life Example

Imagine teaching a child to throw a ball in a basket.

Step 1
Child throws ball → misses.

Step 2
We measure error (distance from basket).

Step 3
We adjust the throw.

Step 4
Child improves.

AI learns exactly the same way.

Summary:

Math	Purpose	Used In AI For
Linear Algebra	Organizing data	Images, text, datasets
Vectors	Represent features	Image pixels, text embeddings
Matrices	Fast calculations	Neural networks
Calculus	Measure error change	Model learning
Derivatives	How error changes	Training models
Gradients	Direction to improve	Gradient descent
Optimization	Reduce error	Improve accuracy

One Sentence to Remember

- Linear Algebra stores the data.
- Calculus tells how to improve.
- Optimization makes the model better.

Conversion of Image into Vector:

[3]:
```
#code to convert image into vector
import numpy as np
from PIL import Image
img = Image.open("C:\\Users\\atir.raza\\Downloads\\cat.jfif")
#apni image ko array mai convert kraingy using numpy operation

img_arr = np.array(img)

#convert matrix into vector now using flatten
'''Flatten ka matlab hota hai multi-dimensional data (matrix ya image) ko ek single line list (vector) mein convert karna.

Simple words mein:

Table ko seedhi line bana dena = Flatten'''

# we will simple call vector as

vector = img_arr.flatten()
#print kraingy results

print("image shape", img_arr.shape)
print("First 20 numbers of vector:", vector[:20])
print("Total numbers in vector:", len(vector))
```

```
image shape (159, 318, 3)
First 20 numbers of vector: [162 165 170 162 165 170 162 165 170 162 165 170 162 165 170 162 165 170
 162 165]
Total numbers in vector: 151686
```

Vector:

[4]:
```
#vectors are basically linear algebra
#vectors mean simple list of numbers which we have already done with it.
import numpy as np   # numpy library import kar rahe hain

# aik simple vector banate hain
vector = np.array([10, 20, 30, 40])

# vector print karte hain
print("Vector:", vector)

# vector ka size check karte hain
print("Vector length:", len(vector))
```

```
Vector: [10 20 30 40]
Vector length: 4
```

Matrix:

[5]:
```
# Matrix example
# matrix numbers ka table hota hai (rows aur columns)

import numpy as np

# simple matrix create karte hain
matrix = np.array([[1,2,3],
                   [4,5,6]])

# matrix print karte hain
print("Matrix:")
print(matrix)

# matrix ka shape dekhte hain
print("Matrix shape:", matrix.shape)
```

```
Matrix:
[[1 2 3]
 [4 5 6]]
Matrix shape: (2, 3)
```

Flatten Example (Matrix → Vector)

```
# Flatten Example (Matrix → Vector)
# Flatten example
# flatten matrix ko aik single line vector bana deta hai

import numpy as np

matrix = np.array([[1,2,3],
                   [4,5,6]])

# matrix ko vector mai convert karna
vector = matrix.flatten()

print("Original Matrix:")
print(matrix)

print("Flatten Vector:")
print(vector)
```

```
Original Matrix:
[[1 2 3]
 [4 5 6]]
Flatten Vector:
[1 2 3 4 5 6]
```

Simple Derivative Idea (Calculus)

```
#Simple Derivative Idea (Calculus)

# derivative ka concept
# derivative change ko measure karta hai

import numpy as np

x = np.array([1,2,3,4,5])   # x values

y = x**2   # y = x^2 function

print("x values:", x)
print("y values:", y)

"""jab x change hota hai to y bhi change hota hai
derivative batata hai change kitni speed sy ho raha hai"""
```

```
x values: [1 2 3 4 5]
y values: [ 1  4  9 16 25]
'jab x change hota hai to y bhi change hota hai\nderivative batata hai change kitni speed sy ho raha hai'
```

Gradient:

[9]:
```
# gradient ka matlab direction of change

import numpy as np

# simple error values
error = np.array([10,8,6,4,2])

print("Error values:", error)

print("Dekhein error slowly slowly kam ho raha hai")

#Important point:

#Yeh error automatically kam nahi ho raha.
#Humne khud numbers aise likh diye hain.
```

Basic Optimization Idea

[10]:
```
# optimization example
# yahan hum error ko kam karny ki koshish karte hain

error = 20

print("Initial Error:", error)

# error ko dheere dheere kam karte hain
error = error - 5

print("New Error:", error)
```

```
Initial Error: 20
New Error: 15
```

Mini AI Learning Simulation

[11]:
```
# AI learning ki simple simulation

prediction = 70
actual = 90

# error calculate karte hain
error = actual - prediction

print("Prediction:", prediction)
print("Actual:", actual)
print("Error:", error)

"""AI prediction karti hai
phir error calculate karti hai
phir apne model ko improve karti hai"""
```

```
Prediction: 70
Actual: 90
Error: 20
```

Tasks:

Task 6: Create a **vector of AI marks for 5 students**.Then print the **average marks** of the class.

Task 7: Create a **vector of pixel brightness values** and find:

- Brightest pixel
- Darkest pixel

Task 8: Create a **3×3 matrix** representing scores of students in three subjects.
Print the matrix and its **shape**.

Task 9: Create a **2×3 matrix** and convert it into a **vector using flatten()**.

Task 10: Suppose an AI predicts the **house price as 80** but the **actual price is 100**.
Calculate the **error**.

Task 11:

Actual value is **50**. Two predictions are **40** and **48**. Calculate error for both and find **which prediction is better**. (Hint: use Abs (absolute function)

Task 12:

Generate **10 random numbers** between 0 and 100 and find:

- Maximum value
- Minimum value

Use data = np.random.randint(0,100,10) to generate random numbers

Task 13:

Suppose AI model **error is 20.**
Reduce the error by **5** and print the new error.

Task 14:

Create a vector of marks and find the **highest score**.

Task 15: Using the some marks vector, find the **index (position) of the top student**.(use argmax)

Task 16: Plot a line graph for numbers: `1,2,3,4` and values: `10,20,25,30`.

Task 17: Plot the same line graph but change the color to **red**.

Task 18: Create a bar chart of student marks.

Students: `Ali, Sara, John`
Marks: `80, 90, 70`

Task 19 Show relationship between study hours and marks(Scatter Ploter).

Hours: `1,2,3,4`
Marks: `50,60,70,80`

Task 20: Show relationship between study hours and marks.

Hours: `1,2,3,4`
Marks: `50,60,70,80`

Task 21 : Show percentage of students in three courses through pie chart.

Courses: `AI, Data Science, Cyber Security`
Students: `40,35,25`

Sample Solution:

```python
# Task 6 Solution
# vector numbers ki simple list hoti hai

import numpy as np

# 5 students ke marks
marks = np.array([70, 85, 90, 65, 88])

# marks print karna
print("Student Marks:", marks)

# average calculate karna
print("Average Marks:", np.mean(marks))
```

```python
# Task 7 Solution
# pixel values ko vector ki form mai store kar rahe hain

import numpy as np

pixels = np.array([120, 200, 90, 45, 255])

print("Pixel Values:", pixels)

# sab se bright pixel
print("Brightest Pixel:", np.max(pixels))

# sab se dark pixel
print("Darkest Pixel:", np.min(pixels))
```

```python
# Task 8 Solution
# matrix numbers ka table hota hai

import numpy as np

scores = np.array([[80,70,90],
                   [60,75,85],
                   [90,88,92]])

print("Scores Matrix:")
print(scores)

# matrix ka size dekhna
print("Matrix Shape:", scores.shape)
```

```python
# Task 9 Solution
# flatten matrix ko ek line vector mai convert karta hai

import numpy as np

matrix = np.array([[1,2,3],
                   [4,5,6]])

# flatten operation
vector = matrix.flatten()

print("Matrix:")
print(matrix)

print("Vector after flatten:")
print(vector)
```

```python
# Task 10 Solution
# prediction aur actual value ka difference error hota hai

actual_price = 100
prediction = 80

error = actual_price - prediction

print("Actual Price:", actual_price)
print("Prediction:", prediction)
print("Error:", error)
```

```python
# Task 11 Solution

actual = 50

prediction1 = 40
prediction2 = 48

# error calculate karna
error1 = abs(actual - prediction1)
error2 = abs(actual - prediction2)

print("Error of Prediction 1:", error1)
print("Error of Prediction 2:", error2)
```

```python
# Task 12 Solution
# random data generate karna

import numpy as np

data = np.random.randint(0,100,10)

print("Random Data:", data)

print("Maximum Value:", np.max(data))
print("Minimum Value:", np.min(data))
```

```python
# Task 13 Solution
# AI learning simulation

error = 20

print("Initial Error:", error)

# model improve hua
error = error - 5

print("New Error:", error)
```

```python
# Task 14 Solution

import numpy as np

marks = np.array([65, 80, 90, 75, 85])

print("Marks:", marks)

# highest score
top_score = np.max(marks)

print("Top Score:", top_score)
```

```python
# Task 15 Solution

import numpy as np

marks = np.array([65, 80, 90, 75, 85])

# highest score ki position
index = np.argmax(marks)

print("Marks:", marks)
print("Top Student Index:", index)
```

```python
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [10,20,25,30]

plt.plot(x,y)
plt.title("Simple Line Graph")
plt.show()
```

```python
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [10,20,25,30]

plt.plot(x,y,color="red")
plt.title("Red Line Graph")
plt.show()
```

```python
import matplotlib.pyplot as plt

students = ["Ali","Sara","John"]
marks = [80,90,70]

plt.bar(students,marks)
plt.title("Student Marks")
plt.show()
```

```python
import matplotlib.pyplot as plt

hours = [1,2,3,4]
marks = [50,60,70,80]

plt.scatter(hours,marks)
plt.title("Study Hours vs Marks")
plt.show()
```

```python
import matplotlib.pyplot as plt

scores = [50,55,60,60,65,70,75,80,85]

plt.hist(scores)
plt.title("Score Distribution")
plt.show()
```

```python
import matplotlib.pyplot as plt

courses = ["AI","Data Science","Cyber Security"]
students = [40,35,25]

plt.pie(students,labels=courses,autopct="%1.1f%%")
plt.title("Course Distribution")
plt.show()
```